Project1: Interim Data Analysis Plan

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Introduction

Highly Active Antiretroviral Therapy (HAART) is the standard treatment regimen for patients diagnosed with HIV. HAART suppresses viral replication, improving immune function as reflected by increases in CD4+ T cell counts, and substantially reducing mortality and disease progression. The Multicenter AIDS Cohort Study is an ongoing prospective cohort study of the natural and treated histories of HIV-1 infection in homosexual and bisexual men in four major cities in the United States. The dataset used in this project includes annual longitudinal laboratory and quality of life measurements, along with demographic and clinical health information, collected for up to eight years after HIV-infected men initiated HAART.

Investigators are interested in conducting a secondary data analysis to evaluate whether hard drug use at baseline is associated with poorer treatment response following HAART initiation. The primary scientific hypothesis is that subjects who report hard drug use at baseline will have worse treatment response two years after initiating HAART compared to subjects who do not report hard drug use. Treatment response will be evaluated using four complementary outcomes: viral load, CD4+ T cell count, and physical and mental quality of life scores.

The corresponding statistical hypotheses evaluate whether mean treatment response at two years differs by baseline hard drug use status for each outcome, adjusting for age, BMI, smoking status, education, and race/ethnicity. For each outcome, the null hypothesis is that there is no difference in mean treatment response between subjects who report hard drug use at baseline and those who do not, while the alternative hypothesis is that a difference exists. These hypotheses will be evaluated using both Bayesian and frequentist methods.

Preliminary Methods

The dataset includes 715 unique patients. The primary exposure variable (baseline hard drug use) and several adjustment variables (race/ethnicity, education, smoking status, and treatment adherence) were recorded numerically in the raw dataset but represent categorical variables. These variables were converted to factor variables with descriptive labels to improve interpretability in descriptive summaries and modeling.

Because the primary research question focuses on treatment response between baseline (year 0) and two years after HAART initiation (year 2), the analysis dataset was created by restricting observations to years 0 and 2 and restructuring the data to create patient-level outcomes. Change-from-baseline outcomes were constructed for CD4+ T cell count and physical and mental quality of life scores as the difference between year 2 and year 0 values. Due to skewness in viral load change, the natural logarithm of the ratio of viral load at year 2 to baseline viral load was used instead. These outcomes allow estimation of average treatment response differences between exposure groups using regression-based methods.

Primary analyses will evaluate the association between baseline hard drug use and treatment response using multivariable linear regression models for each outcome, adjusting for age, BMI, smoking status, education, and race/ethnicity. Both frequentist and Bayesian linear regression models will be fit for each outcome to allow comparison of inference across frameworks. Frequentist results will be summarized using estimated coefficients, standard errors, 95% confidence intervals, and p-values. Bayesian results will be summarized using posterior means, 95% credible intervals, and posterior probabilities that regression coefficients differ from zero.

Additionally, mediation analyses will be conducted using linear regression-based approaches to evaluate whether treatment adherence mediates the relationship between baseline hard drug use and treatment response. Both Bayesian and frequentist mediation models will be fit and compared using corresponding interval estimates and measures of statistical or posterior evidence.

Preliminary Analysis

Table 1: Baseline characteristics stratified by baseline hard drug use.

Variable	Level	Overall	No hard drugs	Yes hard drugs
Sample size	N	715	649	66
BMI (baseline)	Mean (SD)	34.8 (94.4)	36.1 (99.2)	22.4 (6.8)
	Median [IQR]	24.5 [22.1, 27.7]	24.7 [22.4, 27.7]	22.3 [19.9, 25.7]
	Missing	27 (3.8%)	27 (4.2%)	0 (0.0%)
Age (baseline)	Mean (SD)	42.6 (9.4)	42.4 (9.4)	43.7 (9.9)
	Median [IQR]	42.0 [36.0, 49.0]	42.0 [36.0, 49.0]	45.5 [35.2, 52.0]
	Missing	0 (0.0%)	0 (0.0%)	0 (0.0%)
Smoking status (baseline)	Never smoked	188 (26.3%)	187 (28.8%)	1 (1.5%)
	Former smoker	223 (31.2%)	211 (32.5%)	12 (18.2%)
	Current smoker	304 (42.5%)	251 (38.7%)	53 (80.3%)
	Missing	0 (0.0%)	0 (0.0%)	0 (0.0%)
Adherence category (year 2)	100%	211 (41.7%)	190 (40.7%)	21 (53.8%)
	95-99%	243 (48.0%)	226 (48.4%)	17 (43.6%)
	75-94%	39 (7.7%)	38 (8.1%)	1 (2.6%)
	<75%	13 (2.6%)	13 (2.8%)	NA
	Missing	209 (29.2%)	182 (28.0%)	27 (40.9%)
Race/ethnicity (baseline)	White, non-Hispanic	422 (59.0%)	390 (60.1%)	32 (48.5%)
	White, Hispanic	29 (4.1%)	29 (4.5%)	0 (0.0%)
	Black, non-Hispanic	208 (29.1%)	180 (27.7%)	28 (42.4%)
	Black, Hispanic	9 (1.3%)	9 (1.4%)	0 (0.0%)
	Other	8 (1.1%)	8 (1.2%)	0 (0.0%)

Table 1: Baseline characteristics stratified by baseline hard drug use. *(continued)*

Variable	Level	Overall	No hard drugs	Yes hard drugs
	Other Hispanic	39 (5.5%)	33 (5.1%)	6 (9.1%)
	Missing	0 (0.0%)	0 (0.0%)	0 (0.0%)
Education (baseline)	8th grade or less	5 (0.7%)	5 (0.8%)	0 (0.0%)
	9, 10, or 11th grade	50 (7.0%)	30 (4.6%)	20 (30.3%)
	12th grade	125 (17.5%)	112 (17.3%)	13 (19.7%)
	At least one year college but no degree	254 (35.5%)	235 (36.2%)	19 (28.8%)
	Four years college/ got degree	140 (19.6%)	136 (21.0%)	4 (6.1%)
	Some graduate work	50 (7.0%)	50 (7.7%)	0 (0.0%)
	Post-graduate degree	91 (12.7%)	81 (12.5%)	10 (15.2%)
	Missing	0 (0.0%)	0 (0.0%)	0 (0.0%)